Student Performance Analysis System

Project Documentation – Assessment Activity

1. Project Title

Project Name: Student Performance Analysis System

Detail	Information
Student Name	Muneeb Bashir
Course / Semester	B.Tech, 6th Semester
Department	CSE
Institute	Government College of Engineering & Technology, Safapora, Ganderbal
Submitted To	Chandini Madam
Date	19/10/2026

2. Objective

In most colleges, student marks are stored in files or registers, and working out totals, averages, grades and pass/fail results by hand takes a lot of time and can easily go wrong. It is also hard to quickly see who the toppers are or which subject the class is weak in.

This project was created to solve that problem. It reads student data from a CSV file and automatically calculates total marks, average marks, grade and result for every student. It also gives a class summary, a subject-wise analysis and a list of top-performing students. The aim was to practise using Python, NumPy and Pandas on a real, practical problem.

3. Technologies Used

Tool	Purpose
Python	Main programming language

Tool	Purpose
Pandas	Reading the CSV file, working with the data table, sorting, filtering and saving output
NumPy	Numerical calculations on the marks (used in the functions file)
CSV file	Input data and saved output
Git and GitHub	Version control and project submission
VS Code	Writing and running the code

4. Dataset Description

The project uses a file called `students.csv`, which contains the marks and details of students.

- **Approximate number of records:** 15 students
- **Important columns:**

Column	Description
Student_ID	Unique ID of each student
Name	Name of the student
Department	Department the student belongs to
Python, DBMS, Networking, AI	Marks obtained in each subject
Attendance	Attendance percentage of the student

After processing, the program adds these columns: **Total_Marks, Average_Marks, Grade and Result.**

5. Implementation

The project has two Python files. `main.py` controls the flow of the program, and `functions.py` holds the reusable functions.

Step 1 – Loading the data. The program first checks that `students.csv` exists in the same folder. If not, it prints an error message and stops. If the file is there, it is read into a Pandas DataFrame using `pd.read_csv()`. The number of students, the column names and the first five records are printed to check the data.

Step 2 – Processing the marks. The function `calculate_totals_and_averages()` adds the Total_Marks and Average_Marks columns for each student. Then `assign_grade()` is applied on the average marks to create the Grade column, and `assign_result()` is applied row by row (using `axis=1`) to decide Pass or Fail. The grading rule is based on the average marks: 90 and above is A+, 80 and above is A, 70 and above is B, 60 and above is C, 50 and above is D, and below 50 is F. A student passes only if the marks in every subject are 40 or more and attendance is 40% or more. Otherwise the result is Fail.

Step 3 – Student-wise results. The data is sorted by average marks from highest to lowest, and the ID, name, department, total, average, grade, attendance and result of every student are printed.

Step 4 – Class summary. Using Pandas, the program finds the class average, the highest scorer (`idxmax()`) and the lowest scorer (`idxmin()`). It also counts how many students passed and failed and works out the pass percentage.

Step 5 – Subject-wise analysis. The function `subject_analysis()` calculates the performance of each subject and returns it as a new DataFrame. The subject with the highest class average is shown as the best-performing subject.

Step 6 – Top performers. Students whose average marks are 80 or more are filtered out and displayed in order of average marks.

Step 7 – Saving the output. The final results are saved into two files, `student_results.csv` and `subject_analysis.csv`.

6. Key Features

- Loads student data automatically from a CSV file.
- Shows an error message if the data file is missing.
- Calculates total and average marks for each student.
- Assigns a grade and a Pass/Fail result.
- Ranks students from highest to lowest average.
- Gives a class summary: class average, topper, lowest scorer, pass and fail count, pass percentage.

- Does a subject-wise analysis and finds the best subject.
- Lists top-performing students (average of 80 or above).
- Saves the results in two new CSV files.
- Uses separate reusable functions so the code stays clean and easy to change.

7. Output / Screenshots

```
============================================================
STUDENT PERFORMANCE ANALYSIS
============================================================
Total students: 15
Columns: ['Student_ID', 'Name', 'Department', 'Python', 'DBMS', 'Networking', 'AI', 'Attendance']

 Student_ID         Name Department  Python  DBMS  Networking  AI  Attendance
       2301   Divya Nair        CSE      91    81          88  86          97
       2302   Zainab Mir        ECE      67    87          39  94          91
       2303  Pooja Raina        ECE      95    72          69  75          90
       2304 Rahul Pandit        CSE      76    44          88  80          95
       2305    Imran Dar         IT      54    61          58  63          55

============================================================
STUDENT WISE RESULTS
============================================================
 Student_ID           Name Department  Total_Marks  Average_Marks Grade  Attendance Result
       2306     Owais Lone        ECE        365.0          91.25    A+          99   Pass
       2312  Adnan Hussain        CSE        365.0          91.25    A+          60   Pass
       2314    Arjun Mehta         IT        352.0          88.00     A          82   Pass
       2308    Neha Kapoor        ECE        348.0          87.00     A          65   Pass
       2301     Divya Nair        CSE        346.0          86.50     A          97   Pass
       2303    Pooja Raina        ECE        311.0          77.75     B          90   Pass
       2307 Sumaiya Rather         IT        299.0          74.75     B          94   Pass
       2315 Bushra Zahgeer        ECE        290.0          72.50     B          96   Pass
       2309  Muneeb Bashir        ECE        290.0          72.50     B          98   Pass
       2304   Rahul Pandit        CSE        288.0          72.00     B          95   Pass
       2302     Zainab Mir        ECE        287.0          71.75     B          91   Fail
       2311     Nusrat Jan        ECE        255.0          63.75     C          88   Pass
       2310   Faizan Malik        CSE        239.0          59.75     D          92   Fail
       2305      Imran Dar         IT        236.0          59.00     D          55   Pass
       2313 Karan Malhotra         IT        201.0          50.25     D          70   Pass

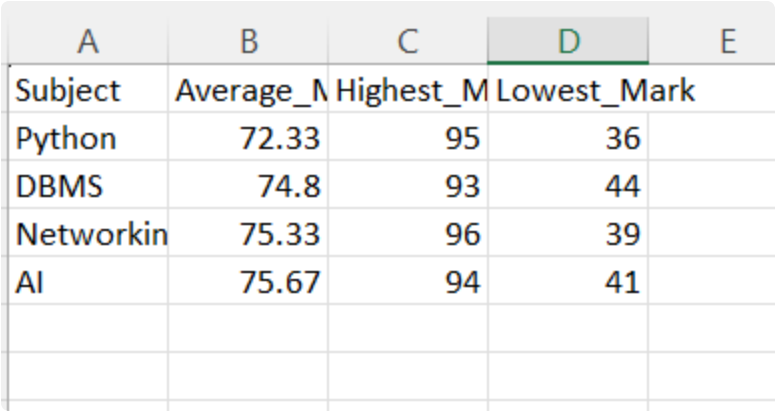
============================================================
```

```
============================================================
CLASS SUMMARY
============================================================
Class average: 74.53
Highest: Owais Lone - 91.25
Lowest: Karan Malhotra - 50.25
Passed: 13
Failed: 2
Pass percentage: 86.67 %

============================================================
SUBJECT WISE PERFORMANCE
============================================================
   Subject  Average_Marks  Highest_Mark  Lowest_Mark
    Python          72.33            95           36
      DBMS          74.80            93           44
Networking          75.33            96           39
        AI          75.67            94           41
Best subject (by class average): AI

============================================================
TOP STUDENTS (average >= 80)
============================================================
 Student_ID          Name  Average_Marks Grade
       2306    Owais Lone          91.25    A+
       2312 Adnan Hussain          91.25    A+
       2314   Arjun Mehta          88.00     A
       2308   Neha Kapoor          87.00     A
       2301    Divya Nair          86.50     A
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Student_ID	Name	Departmer	Python	DBMS	Networkin	AI	Attendanc	Total_Mar	Average_M	Grade	Result		
2301	Divya Nair	CSE	91	81	88	86	97	346	86.5	A	Pass		
2302	Zainab Mir	ECE	67	87	39	94	91	287	71.75	B	Fail		
2303	Pooja Rain	ECE	95	72	69	75	90	311	77.75	B	Pass		
2304	Rahul Panc	CSE	76	44	88	80	95	288	72	B	Pass		
2305	Imran Dar	IT	54	61	58	63	55	236	59	D	Pass		
2306	Owais Lon	ECE	85	93	96	91	99	365	91.25	A+	Pass		
2307	Sumaiya R	IT	79	48	84	88	94	299	74.75	B	Pass		
2308	Neha Kapc	ECE	88	84	86	90	65	348	87	A	Pass		
2309	Muneeb Bi	ECE	86	69	51	84	98	290	72.5	B	Pass		
2310	Faizan Mal	CSE	36	78	79	46	92	239	59.75	D	Fail		
2311	Nusrat Jan	ECE	42	83	89	41	88	255	63.75	C	Pass		
2312	Adnan Hus	CSE	92	90	94	89	60	365	91.25	A+	Pass		
2313	Karan Mall	IT	48	52	46	55	70	201	50.25	D	Pass		
2314	Arjun Meh	IT	82	91	92	87	82	352	88	A	Pass		
2315	Bushra Zal	ECE	64	89	71	66	96	290	72.5	B	Pass		



A	B	C	D	E
Subject	Average_M	Highest_M	Lowest_Mark	
Python	72.33	95	36	
DBMS	74.8	93	44	
Networkin	75.33	96	39	
AI	75.67	94	41	

8. Final Outcome

The project works as a complete system for analysing student results. When `main.py` is run, it reads the data, calculates all the marks, grades and results, and prints a clear report on screen. It also saves the processed data in CSV files that can be opened in Excel.

It gives useful information such as who the toppers are, how many students passed or failed, and which subject the class performs best in. What would take a long time by hand is done in a few seconds. For the 15 students in the dataset, the class average came out to 74.53 and the pass percentage was 86.67% (13 passed and 2 failed). The highest average was 91.25 (Owais Lone and Adnan Hussain, both with grade A+), and the lowest was 50.25. Five students scored an average of 80 or more, and AI was the best-performing subject with a class average of 75.67.

9. Challenges & Learning

Edit this section to match the problems you actually faced.

Challenge	How it was solved
The program crashed when the CSV file was not in the same folder	Added a check for the file path with an error message before loading the data
Applying the result logic needed more than one column at a time	Used `apply()` with `axis=1` so the function works on a whole row
Finding the topper and the lowest scorer	Learned to use `idxmax()` and `idxmin()` with `loc`

Challenge	How it was solved
Keeping the code organised	Moved calculations into a separate `functions.py` file and called them from `main.py`

What I learned: how to load and handle data with Pandas, how to use NumPy for calculations, how to sort and filter a DataFrame, how to write reusable functions, how to split a project into files, and how to upload a project to GitHub.

10. GitHub Repository

Repository link: https://github.com/ZargarMuneeb/student-performance-analysis